TRACE: Memory in Motion

In-Context Learning with Recurrent Memory

AUTHORITATIVE TECHNICAL CONCEPT BRIEFING | DATAFORGE 2026 × PATHWAY TRACK (PS1)

The Central Falsifiable Claim: A fixed-size recurrent state can absorb an unbounded sequence of key–value demonstrations without growing in memory footprint. However, when demonstrations share overlapping (non-orthogonal) keys, readout for one leaks the value of the other — a deterministic, mathematically provable form of forgetting caused by bounded geometric capacity, not an arbitrary scripted failure. Increasing state dimension *d* expands the orthogonal basis, directly eliminating interference.

1. Motivation: The Transformer Context Bottleneck

Standard Transformer LLMs process in-context demonstrations by appending representations to an uncompressed Key-Value (KV) cache. This creates severe design pressures: (1) **Unbounded Memory & Latency:** KV memory scales linearly $O(T)$ with sequence length, exhausting GPU SRAM/HBM limits and increasing generation cost per token; (2) **Externalized Reasoning:** Transformers cannot compress demonstrations into persistent internal state; they must emit long "Chain-of-Thought" (CoT) text tokens to perform multi-step deduction. **The Recurrent Alternative:** Recurrent fast weights compress sequential demonstrations into an evolving internal state matrix of invariant size ($S \in \mathbb{R}^{d \times d}$). Memory footprint and per-step inference compute remain strictly $O(1)$ with respect to sequence length T.

2. The Computational Substrate

TRACE implements the canonical formulation of recurrent associative fast weights:

- **Recurrent State:** $S_t \in \mathbb{R}^{d \times d}$, initialized to $S_0 = \mathbf{0}_{d \times d}$. Memory footprint is strictly d^2 scalars, independent of sequence length T.
- **Demonstration Write (Outer-Product Update):** For key $k_t \in \mathbb{R}^d$ and value carrier $v_t \in \mathbb{R}^d$:

$$S_t = S_{t-1} + v_t\, k_t^T$$

- **Normalized Readout:** Given query key $k_q \in \mathbb{R}^d$, the scalar prediction projects S_t onto k_q:

$$\hat{y} = (S_t\, k_q) \;/\; (k_q \cdot k_q)$$

- **Derived Interference Theorem:** Decomposing $S_T = \sum v_i\, k_i^T$, querying key k_a yields:

$$\hat{y}_a = v_a + \sum_{b \neq a} [\, (k_a \cdot k_b) \;/\; (k_a \cdot k_a)\,]\; v_b$$

  When keys are mutually orthogonal ($k_a \cdot k_b = 0$), cross-talk is zero. When keys overlap ($k_a \cdot k_b \neq 0$), memory leakage is governed entirely by the cosine projection, proving that forgetting is geometric rather than stochastic.

3. Rigorous Architectural Comparison

Dimension	Standard Transformer (KV Cache)	Linear Transformers / Fast Weights	BDH-CQ (Demonstration Variant)	TRACE (Educational Toy Model)
Inference State	Expanding $O(T)$ tokens	Constant $O(d^2)$ matrix	Constant $O(d^2)$ state S	Constant $O(d^2)$ matrix
Read Cost / Step	$O(T)$ attention	$O(d^2)$ matrix-vector product	$O(d^2)$ recurrent update	$O(d^2)$ matrix-vector product
Adaptation Rule	Context concatenation	Linear associative sum	Learned $S_t = U_\theta(S_{t-1}, D_t)$	Rank-1 write $S_t = S_{t-1} + v_t\, k_t^T$
Capacity	Bounded by GPU RAM	Bounded by dimension *d*	Bounded by state & gating	Bounded by dimension *d* (≤ *d* orthogonal keys)
Interference	Zero (exact dictionary)	Linear geometric overlap	Controlled via forget gates	Exact leakage: $\sum_{b \neq a} [(k_a \cdot k_b)/(k_a \cdot k_a)]\, v_b$
Interpretability	Attention token weights	Associative matrix	Black-box learned U_θ	100% transparent linear algebra

4. Literature Grounding & Honest Scope Boundary

BDH-CQ (Engdahl, Kosowski, Chorowski, Stamirowska et al., 2026 — arXiv:2608.09888): §3.2 formalizes demonstration adaptation as a recurrent state update $S_t = U_\theta(S_{t-1}, D_t)$, citing the linear additive update $S_t = S_{t-1} + U_\theta(D_t)$ as the linear fast-weight representation. BDH-CQ proves recurrent state can solve symbolic reasoning benchmarks (e.g. ARC-AGI) without written chain-of-thought tokens.
The Dragon Hatchling / BDH (Kosowski et al., 2025 — arXiv:2509.26507): Formulates self-attention as fast Hebbian synaptic plasticity, treating inference memory as an active biological conductance state.
Mamba / State Spaces (Gu & Dao, 2023 — arXiv:2312.00752): Analyzes theoretical compression limits within fixed recurrent dimensions.
Pre-2022 Foundations: Schlag et al. (ICML 2021) and Katharopoulos et al. (ICML 2020) established outer-product fast weight programming and recurrent linear attention.

Explicit Non-Claim: TRACE is an educational toy model instantiating the additive special case named in BDH-CQ §3.2. TRACE does *not* replicate BDH-CQ's proprietary learned U_θ, deep non-linear gating, ~5% biological activation sparsity, or ARC-AGI benchmark weights.

5. Empirical Verification & Evidence

- **Lossless Recall (d=3):** Absorbing `red→1.0, blue→2.0, green→3.0` into $S_3 \in \mathbb{R}^{3\times3}$. Constructing orthonormal keys via QR decomposition ($k_i \cdot k_j = 0$) guarantees zero cross-talk; query `green` returns **3.000** with **0.000 error**.
- **Geometric Interference (d=2):** Demonstrating ($k_A \rightarrow 1.0$) and ($k_B \rightarrow 9.0$) with overlap $k_A \cdot k_B = 0.60$. Readout $\hat{y}_A = 1.0 + 0.60 \times 9.0 =$ **6.400**, matching theoretical leakage (+5.400) to float64 machine precision.
- **Capacity Phase Transition ($d \in [2, 6]$):** For N=4 demonstrations, error is substantial at d=2 (MSE = 3.13), but drops to machine precision (1.26×10^{-31}) at $d \geq 4$ as an orthogonal basis becomes available.
- **Dual-Engine Parity:** Python (`core/linear_memory.py`) and TypeScript twin (`web/src/memory/linear_memory.ts`) match with $< 10^{-9}$ tolerance across 10 Python pytest tests and 4 Node.js parity tests. Reproducible notebook: `notebook/recurrent_memory.ipynb` .

6. Limitations & Open Research Questions

Geometric Capacity Limit: A linear matrix in $\mathbb{R}^{d\times d}$ stores at most d orthogonal keys. When $T > d$, interference is mathematically guaranteed.
Absence of Learned Gating: Production models (Mamba, BDH-CQ) mitigate capacity limits via input/forget gates that overwrite stale memories.
The Unanswered Question: In biological models like BDH, does extreme activation sparsity (~5% active) provide near-lossless pseudo-orthogonal capacity in high dimensions without explicit QR orthogonalization?